

MTVF® Module

CLIE™ — Cross-Layer Integrity Engine™

System: MTVF® – Multi-Layer Truth Validation Framework
Category: AI Interpretation Standards
Type: Supervisory Integrity Module
Status: Canonical Module
Version: 1.0

Canonical Definition

Cross-Layer Integrity Engine™ (CLIE™) is a supervisory validation mechanism within the MTVF® framework responsible for monitoring interactions between validation layers and detecting structural inconsistencies, bias propagation, and cross-layer conflicts.

CLIE™ ensures that empirical evidence, consensus verification, and contextual governance remain structurally separated and that decision integrity is preserved across the full validation process.

Integrity exists only when validation layers remain coherent and uncontaminated.

Purpose

CLIE™ supervises the interaction between validation layers within MTVF®.

Its purpose is to ensure that validation processes remain structurally reliable by detecting:

- contradictions between validation layers
- bias propagation between layers
- structural contamination of decision inputs

Without supervisory cross-layer monitoring, multi-layer validation systems risk losing structural integrity.

CLIE™ transforms multi-layer validation into **a measurable and auditable integrity process**

Minimum Implementation Framework (MIF)

The CLIE™ module can be implemented using the following minimum supervisory conditions.

Step 1 — Execute Independent Layer Validation

Each validation layer must produce its output independently.

Layers include:

- empirical validation
 - consensus verification
 - contextual governance assessment
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Step 2 — Lock Validation Outputs

Layer outputs must be recorded and locked before cross-layer comparison to prevent modification or contamination.

Minimum requirement:

- validation record
 - layer identification
 - timestamp
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Step 3 — Perform Cross-Layer Comparison

CLIE™ compares outputs from different validation layers in order to detect structural inconsistencies.

Examples of conflicts include:

- empirical data contradicting consensus signals
 - contextual constraints contradicting operational conclusions
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Step 4 — Classify Integrity State

Detected inconsistencies must be classified using a defined integrity classification level.

Minimum requirement:

- conflict identification
 - integrity classification
 - validation report
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Output

The module produces a **Cross-Layer Integrity Assessment** indicating whether the evaluated decision remains structurally coherent

If cross-layer conflicts exist, the decision state is marked for further validation or review.

Use Case 1

AI-Assisted Financial Risk Assessment

Scenario

A financial institution uses AI models to evaluate credit risk.

The model produces a recommendation based on historical financial data.

CLIE™ Application

Validation layers evaluate:

- empirical financial data
- consensus evaluation from independent risk models
- regulatory lending constraints

CLIE™ compares outputs between layers.

Result:

If empirical model outputs conflict with regulatory constraints or consensus risk indicators, the decision is flagged before execution.

CLIE™ prevents structurally inconsistent financial decisions.

Use Case 2

Automated Decision Systems in Public Governance

Scenario

A public administration deploys automated systems to evaluate eligibility for social programs.

Decisions depend on multiple sources:

- citizen data
- statistical eligibility models
- regulatory program conditions

CLIE™ Application

Validation layers evaluate:

- empirical data accuracy
- consensus verification from independent databases
- contextual legal requirements

CLIE™ supervises cross-layer consistency.

Result:

Decisions are executed only when validation layers remain coherent

Cross-layer conflicts trigger review before final decision approval.

Structural Principle

Reliable decision systems require supervision not only of data but also of the relationships between validation layers.

CLIE™ ensures that validation architecture remains structurally coherent and resistant to bias propagation.

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Canonical Source

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